



Interweaving of self-regulated learning and game-based learning in higher education: a review of academic publications from 2009 to 2020

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Abstract

Researchers have indicated the importance of engaging learners in self-regulated learning (SRL) states when situated in game-based learning contexts; however, it remains a challenge for both educational and educational technology researchers to effectively integrate both. To this end, this study investigated how SRL strategies are interwoven with game-based learning in higher education by searching the web of science database to systematically review the papers published between 2009 and 2020 in academic journals. The encoded dimensions ranged from the primary research purpose to research issues, including application domains, research methods, duration of the studies, SRL strategies, game types, and game genres. It was found that since 2015, the research purposes have become increasingly diverse, with skills acquisition in game-based learning being regarded as the most important goal, followed by knowledge acquisition and behavior change. Such games took goal orientation, peer learning, and regulating as the main SRL strategies, which exerted a positive effect on learning performance, self-efficacy/confidence, attitudes/effort, satisfaction/interest, and learning behavior. Meanwhile, these SRL strategies were well embedded into problem-solving, simulation, multi-type, and RPG game types against the setting of the real-life-related storyline as the main game genre. Since previous studies lacked the systematic application of all SRL strategies within a game-based learning environment, they could not uncover the dynamic and cyclic processes of SRL in game-based learning environments. Hence, this study proposed corresponding suggestions for future research issues as a reference for researchers, teachers, and decision-makers.

Keywords Game-based learning · Higher education · Self-regulated learning (SRL) strategies · Self-regulated learning

Introduction

As games are playful, learners easily engage in entertainment rather than learning when playing games if they lack self-regulated learning (SRL) competence. Self-regulated learning (Corno & Mandinach, 2004), a theory of self-regulation in academic settings, involves motivational, metacognitive, cognitive, affective and behavioral regulations

of learning (Wong et al., 2019), highlighting the reciprocal interaction between person and environment, mediated by behavior (Dinsmore et al., 2008). Self-regulated learning focuses on the interaction among cognitive, motivational and contextual factors. According to previous studies, game-based learning is regarded as a promising way to promote students' motivation and engagement in higher education (Hamari et al., 2016; Troussas et al., 2020), where engagement consists of cognitive, affective and behavioral aspects (Wang et al., 2016). In other words, game-based learning is interwoven with self-regulated learning, which can direct learners towards a specific learning goal, involving self-monitoring behavior (Riemer & Schrader, 2016; Yildiz Durak et al., 2022). In some sense, SRL is crucial for effective game-based learning.

SRL is considered as a key competence for lifelong learning (Dent & Koenka, 2016; Theobald, 2021). In higher education, university students are required to self-direct their own learning and hence SRL is especially important for them (Broadbent, 2017; Broadbent & Poon, 2015). However, SRL does not spontaneously occur. Its occurrence depends to a large degree on social context. Hadwin and Oshige (2011) considered that modeling and prompting are key instructional tools for promoting SRL. Through scrutinizing the literature on serious games in higher education (Gómez & Suárez, 2021), albeit employing a narrative strategy to summarize and synthesize data about the pedagogic, didactic, and technical approaches, no findings were depicted about SRL strategies in serious games. Moreover, to date, some studies have systematically reviewed game-based learning according to its characteristics (Nadolny et al., 2020), and its theoretical basis including goal setting theory, self-efficacy theory, achievement goal theory, and so on, rather than SRL theory (Krath et al., 2021), educational technologies (Hwang & Chen, 2021), and creativity (Behnamnia et al., 2020). These reviews, however, were neither restricted to higher education, nor explored the use of SRL strategies in game-based learning.

According to a literature review synthesized by Cibrian et al. (2022), serious games can implicitly support the development of academic and life skills in home and school settings by encouraging improved self-regulation and collaboration with others. Moreover, Wang et al. (2020) indicated that self-regulation has a negative correlation with addictive behavior. In other words, it is crucial to foster students' self-regulation competence in case of game addiction in game-based learning contexts. Wouters and van Oostendorp (2013) used a meta-analysis method to analyze effects of learning support on learning outcomes, where such learning supports as advice, cues, collaboration, elaboration, and so on were included without referring to SRL strategies. Compared to the SRL framework given by Panadero (2017), most of the research only explored some aspects of SRL strategies in game-based learning, while few covered all aspects of SRL strategies. Hence, in order to systematically clarify the framework of SRL and provide enlightenment regarding how to design SRL strategies in higher education game-based learning for game developers, teachers and researchers, this study investigated the following research questions:

RQ 1: What were the primary research purposes of the research on SRL-supported game-based learning in higher education?

RQ 2: What was the trend of SRL strategies adopted in SRL-supported game-based learning in higher education research?

RQ 3: What game types and game genres were adopted in SRL-supported game-based learning in higher education research?

RQ 4: What were the application domains in the research on SRL-supported game-based learning in higher education?

RQ 5: What were the research methods and duration of these studies in SRL-supported game-based learning in higher education research?

RQ 6: What were the research issues of SRL-supported game-based learning in higher education research?

Literature review

Self-regulated learning strategies in higher education

Self-regulated learning was originally explored five decades ago by the social learning community. There are six models of SRL (Panadero, 2017), of which the Metacognitive and affective model of self-regulated learning (MASRL) presented by Efklides (2011) has been widely adopted. This model involves self-efficacy as one component of motivational aspects in SRL. In 1986, Zimmerman and Martinez-Pons's taxonomy of self-regulated strategies was proposed, consisting of 14 categories. That is (1) self-evaluation, (2) organizing & transforming, (3) goal setting & planning, (4) seeking information, (5) keeping records & monitoring, (6) environmental structuring, (7) self-consequences, (8) rehearsing & memorizing, (9) seeking social assistance from peers, (10) seeking social assistance from teachers, (11) seeking social assistance from adults, (12) rereading tests, (13) rereading notes, and (14) rereading textbooks (Garcia et al., 2018; Zimmerman & Martinez-Pons, 1986). With the evolution of learning environments, self-regulated learning strategies are variously defined by different scholars. Pintrich (2004) defined self-regulated learning strategies as metacognitive, motivational, and behavioral strategies which engage students in monitoring and regulating their own behavior in learning situations in a goal directed way in general. In contrast, Anthonysamy et al. (2020) contended that SRL strategies can be classified into four domains: (1) cognitive engagement, (2) metacognitive knowledge (3) resource management (Zimmerman & Martinez-Pons, 1986) and (4) motivational beliefs (Pintrich, 1999), where cognitive engagement includes rehearsal, elaboration, organization and critical thinking; metacognitive knowledge refers to planning, monitoring and regulating; resource management embraces time and study environment, effort regulation, peer learning, and help seeking; and motivational beliefs consist of self-efficacy, task value and goal orientation, which sum up to 14 categories. With reference to the connotation of regulation in the metacognition domain (Efklides, 2008), it embraces regulation of cognition, emotion and willpower (or effort). Seeking information is necessary for individuals to engage themselves in initiating the assigned task towards a targeted goal, which also belongs to individual cognitive engagement, so in this study 14 categories were adjusted to (1) rehearsal, (2) seeking information, (3) elaboration, (4) organization, (5) critical thinking, (6) planning, (7) monitoring, (8) regulating, (9) time and study environment, (10) peer learning, (11) help-seeking, (12) self-efficacy, (13) task value, and (14) goal orientation; these are slightly different from the 14 categories addressed by Anthonysamy et al. (2020).

These strategies are often assessed with self-report instruments (Roth et al., 2015) such as the *Motivational Strategies for Learning Questionnaire* (MSLQ; Pintrich et al., 1993), the *Learning and Study Strategies Inventory* (LASSI; Weinstein et al., 1987), the *Inventory of Learning Styles* (ILS; Vermunt, 1998), the *Academic Self-Regulated Learning Scale* (A-SRL-S; Magno, 2009), the *Online Self-Regulated Learning Questionnaire* (OSLQ; Barnard et al., 2008), the *Self-Efficacy for Learning Form* (SELF; Zimmerman & Kit-santas, 2007), the *Self-Efficacy for Self-Regulated Learning Scale* (SESRL; Garavalia &

Gredler, 2002; Gredler & Schwartz, 1997), the *Self-Regulated Learning Interview Schedule* (SRLIS; Zimmerman & Martinez-Pons, 1986), and *Thinking aloud* (Azevedo et al., 2004). These instruments involve motivation (e.g., value, expectancy, and affect), metacognition (e.g., rehearsal, elaboration, organization, critical thinking, and metacognitive self-regulation), and behavioral aspects (e.g., time and study environment regulation, effort regulation, peer learning, help seeking, and time management). Moreover, in recent years, sequence mining or process mining technologies have been used to assess self-regulated learning (Saint et al., 2022). Maldonado-Mahauad et al. (2018) used process mining with big data from undergraduate MOOCs to uncover how learning behavior patterns matched self-regulated learning strategies, and identified three clusters of learners: comprehensive, targeting and sampling. Sonnenberg and Bannert (2019) verified the sustainability of metacognitive prompting on self-regulatory behaviors using process mining for undergraduates. Li et al. (2020) used clickstream data to measure sub-factors of self-regulated learning such as time management and effort regulation in online courses for undergraduates.

Game-based learning in higher education

Game-based learning can foster learners' cognitive, behavioral, affective, and sociocultural engagement with the subject matter (Castillo et al., 2023; Plass et al., 2015, 2020). Elements of game design for learning consist of game mechanics (i.e., learning mechanics and assessment mechanics), visual aesthetic design, narrative design, incentive system, musical score, and content & skills. The ways that games can facilitate cognitive processing for learners include situatedness of learning, transfer of learning, scaffolding and relevant feedback, dynamic assessment, information design (i.e., representation of information), interaction design (i.e., learning mechanics), and gestures & movement (Swacha et al., 2023). As for scaffolding, it should be dynamically adaptive in game-based learning, which requires ongoing evaluation of the learner, and fading as learners acquire skills and knowledge. With the sophistication of learning analytics techniques, data mining technology embedded into serious games can be used to assess learning and predict performance, analyze in-game players' behaviors, profile students' characteristics, and analyze the study effect of in-game interventions (Alonso-Fernández et al., 2019). Whether game-based learning has a positive effect on learning outcomes has not reached a consensus (Qin & Clark, 2016). Some studies (Boyle et al., 2014; Nitisakunwut & Hwang, 2023; Vogel et al., 2006) indicated that game-based learning might be superior to traditional classroom instruction as it could increase students' motivation for learning and provide them with opportunities to explore and acquire new knowledge and skills. Nonetheless, other studies (Emes, 1997) found no strong evidence that game-based learning was associated with students' high academic achievements or psychological development. These inconsistent findings regarding the effectiveness of game-based learning might be mediated by learners' engagement. Hamari et al. (2016) explored the impact of flow, engagement and immersion on learning in game-based learning environments and indicated that engagement in the game has a clear positive effect on learning, while immersion in the game did not exert a significant effect on learning. In addition, the challenge of the game has a positive effect on learning both directly and via increasing the engagement, but skill in the game did not affect learning directly but by increasing engagement in the game. Both the challenge of the game and being skilled in the game had positive effects on being both engaged and immersed in the game. In other words, engagement mediated the effect of challenge or skill in the game on learning. Many studies have indicated that some SRL strategies such as self-efficacy (Heo

et al., 2022; Kuo et al., 2021), effort regulation (Heo et al., 2022) and time & environment management (Heo et al., 2022) or a whole set of SRL strategies including cognition, metacognition, social behavior and motivational regulation (Zhou & Hiver, 2022) exerted significant effects on learning engagement, which justified that SRL is crucial for effective game-based learning.

To date, previous literature reviews on game-based learning adopted the meta-analysis method (Clark et al., 2016; Qin & Clark, 2016; Vogel et al., 2006) which focused on the practical significance (i.e., effect size) of empirical studies, and other previous literature reviews utilized the systematic review method, respectively to explore the learning theoretical bases in game-based learning (Wu et al., 2012a), to examine the empirical evidence for the effectiveness of digital games on second language learning (Acquah & Katz, 2020), to examine the effectiveness of digital game-based learning on adult learners' professional upskilling (Tay et al., 2022), to systematically discuss the design of intrinsic integration of domain-specific learning in game mechanics and game world design (Ke, 2015), to explore the game-based learning approach on students' motivation and understanding of chemistry concepts (Byusa et al., 2022), and to identify the theoretical foundations of gamification, serious games and game-based learning (Krath et al., 2021). However, no systematic literature review on SRL interwoven with game-based learning has been conducted in recent years.

Self-regulated learning strategies and game-based learning

According to Zimmerman's (1989) sociocognitive model of self-regulation, social context or environment is an important part of students' SRL. In contrast, game-based learning just aims to afford a playful context to motivate learners to engage in active learning, when SRL easily occurs because contemporary SRL perspectives regard the effects of social and contextual influences on SRL (Corno & Mandinach, 2004). In recent years, a few studies have investigated effects of self-regulated learning strategies on learning performance or achievement in game-based learning environments. Zumbach et al. (2020) indicated that in a serious game, metacognitive and cognitive prompting as one strategy of self-regulated learning proved effective, whereas direct metacognitive training only had a minor effect on knowledge acquisition for high school students. Chen and Hsu (2020) found that self-efficacy, intrinsic value and test anxiety significantly influenced game engagement and game experience when undergraduates used a VR game-based English mobile learning application for learning English. During game-based English learning, self-efficacy affected self-regulation, and self-regulation in the learning environment was at a moderate level. Syal and Nietfeld (2020) utilized self-report measures including achievement goals, situational interest and trace data measuring in-game engagement and disengagement behaviors to verify which approach measuring self-regulated learning could effectively predict performance and content learning. They unveiled that self-reported achievement goals and situational interest were not as effective as in-game trace behaviors in predicting performance and content learning. Taub et al. (2018) used sequence mining to analyze how undergraduates' scientific reasoning along with their metacognitive monitoring of self-regulated learning processes impacted their STEM learning effectiveness in a game-based learning environment. Compared to the self-regulated learning framework given by Panadero (2017), most available research only explored some aspects of self-regulated learning strategies in game-based learning, and few studies covered all aspects of self-regulated learning strategies. Therefore, it is worthwhile to systematically analyze what SRL strategies were used

and on which SRL strategies more attention was focused in game-based learning environments during the recent decade, as it would shed light on future research for game-based learning environment design and guide research avenues for how to effectively use and measure SRL strategies in game-based learning environments.

Research methods

This systematic review and content analysis were conducted according to the rigorous protocols suggested by Moher et al. (2009) and Wu et al. (2012b). In addition, the parts on data sources and data coding and analysis are dissected later.

Process of data searching and collection

The web of science database (WOS) was searched due to its authority and reliability with reference to the previous systematic review papers (Chen et al., 2020; Huang et al., 2022; Lai, 2020). On September 30, 2021, the WOS database was searched for research related to the adoption of game-based learning to facilitate SRL in higher education published until 2020. Table 1 presents the inclusion and exclusion criteria that were established.

Based on Broadbent and Poon (2015), Hwang and Chen (2021), and Lee et al. (2019), this study used the keywords “game-based learning” (“game*” or “gaming” or “gamification”) AND “self-regulated learning” (“self regulat* learning strateg*” or “metacog*” or “learning strateg*” or “self regulat*” or “rehearsal” or “elaboration” or “critical thinking” or “monitoring” or “time management” or “effort regulation” or “peer learning” or “help seeking” or “goal setting” or “environment structur*” or “self efficacy” or “task value” or “planning” or “intrinsic goal orientation” or “extrinsic goal orientation” or “control belief” or “test anxiety” or “task strateg*” or “self evaluat*”) AND “higher education” (“higher education” or “student” or “university” or “postgrad*” or “undergrad*” or “sophomore” or “college” or “course” or “freshman” or “tertiary” or “post-secondary education” or “pupil”), giving 668 papers. The categories were then restricted to “education and educational research,” and the type of literature was limited to articles, based on the suggestions of several previous studies (Hsu et al., 2012; Lai, 2020). This reduced the number of

Table 1 Inclusion and exclusion criteria

Inclusion	Exclusion
a. Participants who were undergraduates, graduates, post-secondary education students, adults, etc	a. Editorials, reviews, correction notes, and early access articles were excluded
b. Research content involving self-regulated learning/SRL, or metacognition, or learning strategies, or critical thinking, or self-efficacy, or engagement, or peer learning, or monitoring, or goal setting, or planning, or test anxiety, or emotional regulation, or cognitive regulation	b. Studies with participants from kindergarten, primary school, middle school, or high school were excluded
c. Learning environments based on serious game or learning contexts gamified only if including one gamification mechanism (e.g., rules, rewards, competition, cooperation, challenges)	c. Empirical studies examining SRL outside of the game-based learning environment were excluded Articles for which the full text was not available were excluded
	d. Conference proceedings were excluded

relevant papers to 269. After manually reviewing the articles that were non-English (e.g., Spanish and Turkish) and not related to the topic of this study, the final number of articles analyzed was 55, as shown in Fig. 1.

Theoretical model and coding schemes

The content coding analysis of the primary research purpose, application domains, research methods and duration of these studies, SRL strategies, game type and genre, and research issues in this study was based on the Technology-based Learning Review model proposed by Hwang and Tu (2021) and Lai (2020). The coding was done by two experienced coders with a kappa score of 0.81 and 5 years of coding experience. Inconsistent coding was discussed until agreement was established to ensure the validity of the coding analysis. For each dimensional code, the description is as follows.

1. Primary research purpose: Referring to Abdul Jabbar and Felicia (2015), the primary research purpose was categorized into (a) games to acquire skills, (b) games to acquire knowledge, (c) games to change behavior, (d) mixed, and (e) others. Here, knowledge acquisition refers to learners actively making sense of content including retention and appreciation of concepts, principles, theorems, laws, etc., while skills acquisition is regarded as application of knowledge that has been learned by means of practicing and processing, acquiring learning skills (analyzing, synthesizing, summarizing, and inferring), subject area skills and problem-solving skills. See Table 5.
2. SRL Strategies: Based on Anthonysamy et al. (2020) and Garcia et al. (2018), the SRL strategies were divided into (a) rehearsal, (b) seeking information, (c) elaboration, (d) organization, (e) critical thinking, (f) planning, (g) monitoring, (h) regulating, (i) time and study environment, (j) peer learning, (k) help-seeking (including online peers, teach-

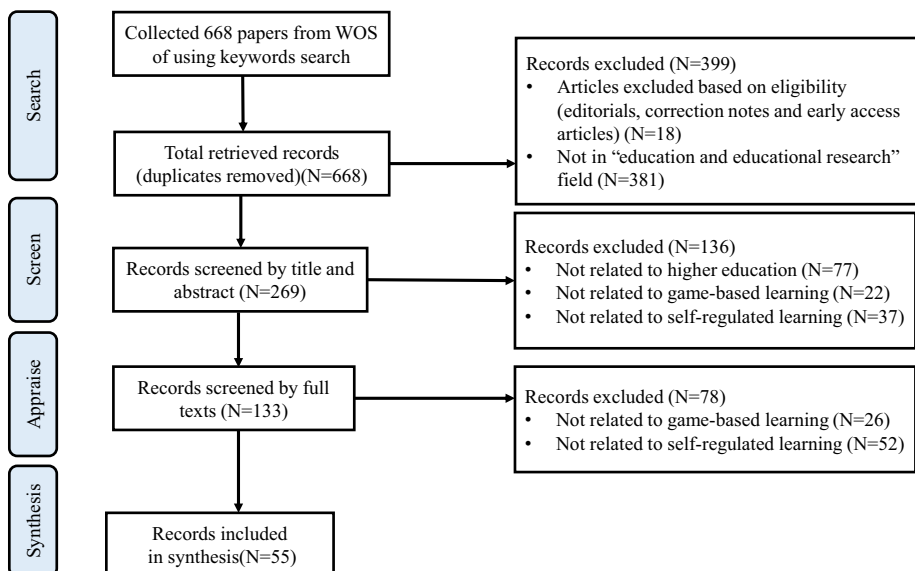


Fig. 1 WOS database searching steps

- ers or others), (l) self-efficacy, (m) task value, (n) goal orientation, and (o) mixed. See the coding scheme in [Table 6](#).
3. Game type and game genre: Referring to Abdul Jabbar and Felicia (2015), game type refers to description of the game-play, and game genre means description of the narrative content of the game. In this study, game type was classified into (a) RPG, (b) puzzle-based, (c) action, (d) competitive, (e) simulation, (f) problem solving, (g) strategy, and (h) multitype. Game genre was divided into (a) fantasy/fiction, (b) real-life related storyline (e.g., home, school), (c) content area related storyline, (d) space exploration, (e) civilization/history, (f) crime, (g) mystery, (h) war, and (i) non-specified. See [Table 7](#).
 4. Application domains: According to Lai (2020), the application domains were divided into (a) science (physics, chemistry, biology), (b) mathematics, (c) arts, (d) language, (e) social studies (including history), (f) engineering (including computer courses), (g) health, medical and physical education, (h) business or management, (i) across-disciplines (e.g., STEM), (j) non-specified, and (k) mixed.
 5. Research methods: According to Lai (2020), the research methods refer to approaches that collect and analyze data for answering the research questions, which were divided into (a) survey, (b) experimental design, (c) qualitative, and (d) mixed. See [Table 8](#).
 6. Duration of these studies: According to Xia and Zhong (2018), the duration of these studies was divided into (a) ≤ 1 day, (b) ≤ 4 weeks, (c) ≤ 8 weeks, (d) ≤ 6 months, (e) ≤ 12 months, (f) > 12 months, and (g) non-specified.
 7. Research issues: With reference to previous studies (Lai, 2020; Tang et al., 2022), the research questions were divided into (a) affect (e.g., technology acceptance model or intention of use, attitudes/efforts, self-efficacy/confidence, satisfaction/interest, engagement, anxiety, cognitive load, and opinions of learners or learning perceptions), (b) cognition (e.g., learning performance, higher order thinking skills, and collaboration or communication), (c) skills (e.g., learners' problem-solving skills, subject area skills, etc.), (d) learning behavior (e.g., log activities, time and efforts for learning activities), (e) correlation or cause-and-effect analysis (e.g., exploration of correlation between variables), and (f) others.

Distribution of publications on SRL-supported game-based learning in higher education by years

Although the publications were retrieved without restriction to the beginning year, according to the literature screening flow shown in [Fig. 1](#), it was found that the pioneer of SRL-supported game-based learning in higher education dates back to 2009, so this study reviewed the publications starting from 2009. [Figure 2](#) documents the growth in publishing of SRL-supported game-based learning in higher education from 2009 to 2020. The earliest paper was written by Sancho et al. (2009), and proposed a framework for applying socially structured educational approaches in higher education to facilitate the formation of complementary and semi-autonomous learning teams that collaborate to solve problems provided by instructors. As suggested by previous review studies (Hsu et al., 2012; Lai, 2020), this study analyzed the number of papers in two periods to better understand the trends of SRL-supported game-based learning in higher education research. A total of seven articles were published in the first period (2009–2014) and 48 in the second period (2015–2020), clearly reflecting the steady interest in applying the framework. Over the past 12 years, the number of SRL-supported game-based learning studies in higher education has steadily increased, and this figure provides a positive

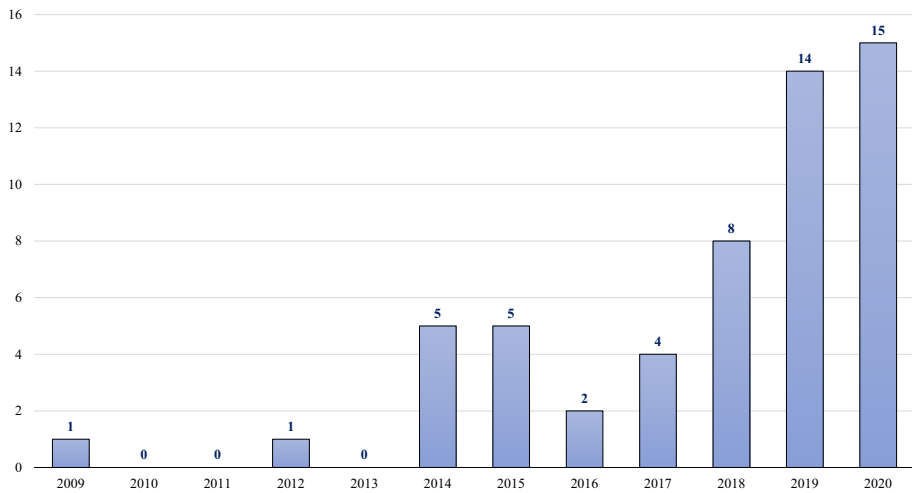


Fig. 2 Trends of SRL-supported game-based learning in higher education by years

indication that there will be more research on SRL-supported game-based learning in higher education in the future.

Research results

Primary research purpose

As reported in Fig. 3, games to acquire skills account for the largest percentage (50.91%) of published articles from 2009 to 2020, followed by games to acquire knowledge (18.18%), mixed (16.36%), and games to change behaviors (10.91%). During the former 6 years (2009–2014), only seven papers were published for using games to acquire knowledge, skills, and both. Moreover, studies on acquiring knowledge, skills, or both were evenly distributed. In contrast, acquiring skills in game-based learning was of increasing interest, and the number of published papers (25) was the greatest during the latter 6 years (2015–2020). Meanwhile, changing behavior in game-based learning also attracted scholars' research interest, with the published papers increasing to six from zero. The number of studies on games to acquire knowledge also increased by six. Studies with more than two purposes (mixed) similarly increased by five articles. In other words, a growing number of studies showed that game-based learning was adopted to gain more skills than knowledge, behavior, or others. Moreover, game-based learning is helpful for comprehensive competence development. For instance, Biswas et al. (2020) used the UrbanSim game to train complex decision-making competence. In addition, Fu et al. (2019) employed a mind mapping-based contextual game approach to significantly improve English writing performance.

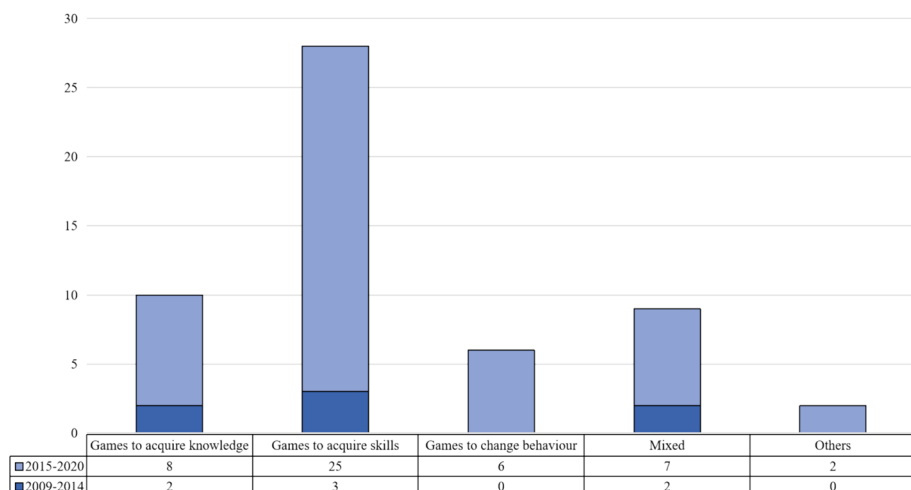


Fig. 3 Distribution of primary research purposes

SRL strategies

As illustrated in Fig. 4, during the period from 2009 to 2014 (7 articles), game-based learning only involved rehearsal (1 article, 14.29%), peer learning (3 articles, 42.86%), and mixed strategies (3 articles, 42.86%). That is, in terms of the number of published articles, less attention was paid to SRL strategies in game-based learning in this period. These studies used game-based practice or rehearsal in writing strategy training (Proske et al., 2014). However, a few SRL strategies are often blended to support learning. These studies usually adopted monitoring and planning strategies integrated into game-based problem-solving

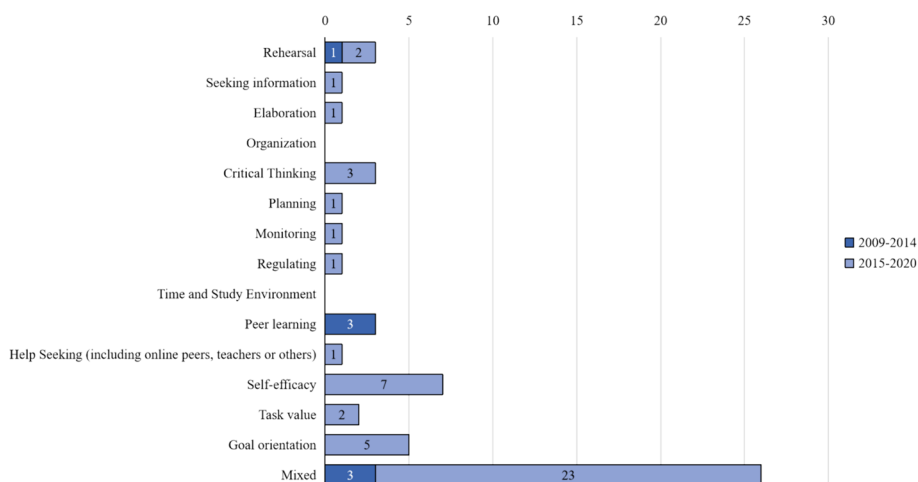


Fig. 4 Distribution of roles of SRL strategies

learning systems to train learners' problem-solving competence (Lin et al., 2014) or interacting with game-oriented tangible user interfaces to learn c# programming (Rodríguez Corral et al., 2014). Besides, Liu and Chen (2012) also used rehearsal, help-seeking, and peer learning strategies to play vocabulary games intended to facilitate lexical growth. In the engineering domain, peer learning or collaborative learning was adopted as an effective strategy to promote complex task learning (González et al., 2014; Venter & Coetzee, 2014).

In the period from 2015 to 2020 (48 articles), SRL strategies attracted increasing attention from scholars. Sub-facets of SRL strategies were differently addressed by studies. Most studies focused their interest on mixed strategies (23 articles, 47.92%) with more than two tactics, where the greatest number of studies involved goal orientation (13 articles, 27.08%), followed by peer learning (9 articles, 18.75%), regulating (9 articles, 18.75%), planning (7 articles, 14.58%), seeking information (7 articles, 14.58%), critical thinking (6 articles, 12.5%), self-efficacy (6 articles, 12.5%), monitoring (5 articles, 10.42%), task value (4 articles, 8.33%), elaboration (2 articles, 4.17%), organization (2 articles, 4.17%), time and study environment (2 articles, 4.17%), rehearsal (1 article, 2.08%), and help-seeking (1 article, 2.08%).

According to categories of SRL strategies (Pintrich, 2004), in the 25 studies other than the mixed strategies studies, the distribution of SRL strategies mainly covered motivational beliefs (i.e., goal orientation, self-efficacy, and task value), metacognitive knowledge (i.e., regulating, planning & monitoring) and cognition engagement (i.e., rehearsal, seeking information, elaboration, organization, critical thinking), and behavioral aspects (i.e., peer learning, time and study environment, help-seeking). In terms of motivational beliefs of SRL strategies, seven studies focused their interest on self-efficacy, five on goal orientation, and two on task value. As far as cognitive & metacognitive strategies are concerned, only three studies attached importance to just critical thinking, there were two studies on rehearsal, and only one for each of seeking information, elaboration, planning, monitoring, and regulating. With behavioral strategies, help-seeking was investigated by only one study. Judging from the holistic distribution in the period of 2015 to 2020 along with the detailed mixed strategies as above, most attention was paid to cognitive & metacognitive strategies (e.g., Lee, 2019; van Ginkel et al., 2019), while motivational strategies (e.g., Rohlfesen et al., 2020; Sanchez et al., 2020) ranked second, and behavioral strategies (e.g., Ibrahim, 2019; Mavridis & Tsiatsos, 2017) ranked third. Although the total number of studies on cognitive & metacognitive strategies was greater than those on motivational and behavioral strategies, sub-facets of cognitive & metacognitive strategies were not studied evenly by researchers. More attention was paid to regulating, planning, and monitoring than to elaboration, organization, seeking information, and rehearsal. For example, Chang et al. (2020) used a problem-based game called "Programmer Adventure Land" to enhance learners' knowledge of computer programming. While playing the game, students experienced peer interaction and gaming context & mission design where they set their own goals to achieve self-directed learning. Meanwhile, the Programmer Adventure Land game provided feedback for students along with tutor's comments, and students' activities were recorded in the game, through which instructors and students could monitor their progress. In such a game-based learning environment, three kinds of SRL strategies, namely goal orientation, peer learning, and monitoring, were embedded. Taub et al. (2020) exploited Crystal Island, a game-based learning environment for microbiology, to engage students in assuming the role of a medical detective to investigate an unidentified outbreak that has spread among a team of scientists at the virtual research station. This game design included four interdependent phases, namely task understanding, setting goals and planning, engaging in

learning strategies, and making adaptations regarding the Information Processing Theory of SRL (Winne, 2018; Winne & Hadwin, 1998, 2008). The four phases are cyclic and take place simultaneously. During each phase, students can experience different self-regulatory processes. Such a game involves such sub-facets of SRL strategies as task value, goal orientation, planning, regulating, and so on. Chen et al. (2019b) developed a game named the My-Hero system with two versions to help students learn the basic concepts of programming. One is the goal-setting version intended to foster students' goal setting and preparation, the other is the negotiation version, the purpose of which is to promote students' regulation, which is classified into goal negotiation and strategy negotiation. Such a game obviously promotes metacognitive learning strategies involving planning and regulation.

Game type and game genre

Figure 5 obviously shows that in game-based learning environments, problem-solving games were the most popular game type (15 articles, 27.27%), followed by simulation games (12 articles, 21.82%) and multi-type games (10 articles, 18.18%) during the years of 2009 to 2020 (55 articles). Although in the period of 2009 to 2014 such game types as above were less but more evenly studied, during the period of 2015 to 2020, these game types sharply increased compared to the corresponding game types of the period of 2009–2014; especially, simulation games increased by 10 articles, followed by multi-type games (8 articles) and RPG (6 articles). In other words, starting from 2015, simulation, problem-solving, and multi-type games have become more preferred game types for learning.

For example, most simulation games applied to business & management (Buil et al., 2019; Khatri et al., 2019; Sholihin, 2020; Tao et al., 2015), and language (Ibrahim, 2019; Lee, 2019; van Ginkel et al., 2019). These games adopted such SRL strategies as self-efficacy, planning, peer learning, seeking information, elaboration, time and study environment, and rehearsal. In a business simulation game, which comprises the game introduction, game practice via simulated competition, group thought-sharing reports, and final

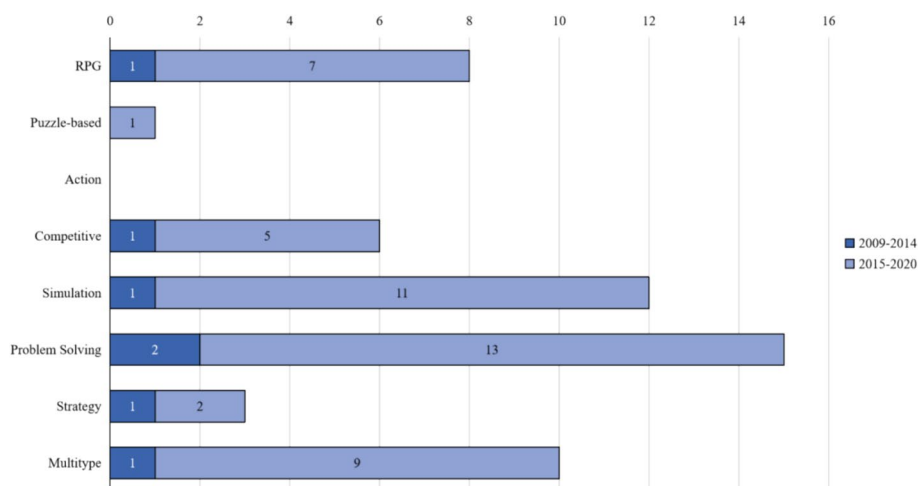


Fig. 5 Distribution of game type

game competition (Tao et al., 2015), students experienced peer collaboration, exploration of useful information, and competitive pressure among students. Khatri et al. (2019) used the competency and career assessment tool (COMPCAT) to engage students in career planning activities with learning analytics features. This game represents a planet to which the students would be taken on a space trip and allowed to spend their vacation with a few friendly aliens. Each alien introduces the student to an area of management and its relevance and contribution to sustaining life on that planet. The major subjects of management, along with their related streams, are introduced to the students: marketing, human resource management, finance, international business, and information technology. The student is initially allowed to stay with all possible selected aliens and then can, in the second phase, make a choice to stay with their preferred aliens. In such a game, students are required to exploit their own planning strategies to accomplish their missions.

As illustrated in Fig. 6, games with real-life-related storylines interwoven with SRL strategies were the most popular game genre (25 articles, 45.45%) during the period of 2009 to 2020. During these years, non-specified game genres ranked second (16 articles, 29.09%), and the content area-related storyline ranked third (9 articles, 16.36%). Compared to the period of 2009 to 2014, when there were only three kinds of game genres such as real-life related storyline, content area related storyline, and non-specified, game genres became more diverse during the years of 2015 to 2020, embracing fantasy/fiction, mystery, and war, as well as the same genres as in the former period.

Table 2 illustrates that mixed SRL strategies were the first most embedded in the real-life related storyline (14 articles, 25.45%), followed by content area related storyline (5 articles, 9.09%) and non-specified (4 articles, 7.27%). Self-efficacy was the SRL strategy that was second most embedded into the real-life-related storyline game genre. For instance, Ding et al. (2018) implemented a gamified online discussion tool named gEchoLu in an undergraduate-level online course. This tool was characterized by gamification, including badges, experience points, leaderboards, progress bar, reactions, and awards. If engaged in an online discussion on this tool, students were asked to meet such needs as follows: (1) clear goals and guidelines, (2) autonomy support, (3) sense of competence, and (4) social

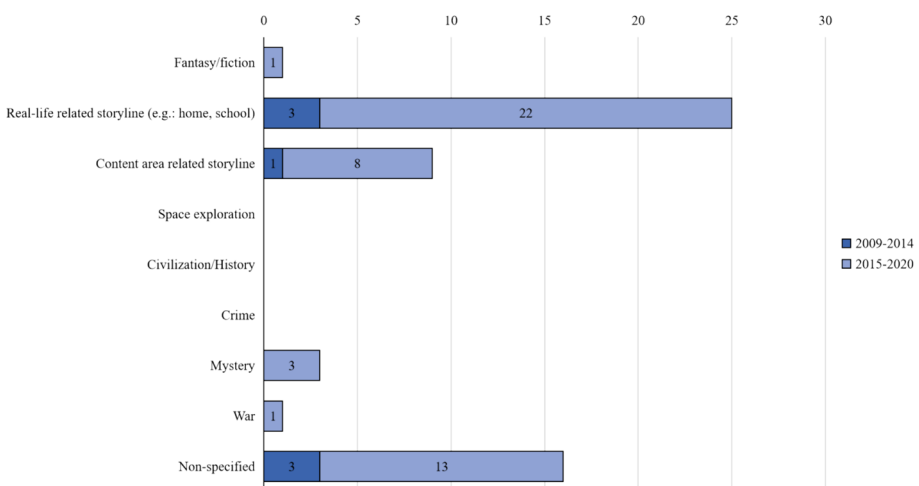


Fig. 6 Distribution of game genres

Table 2 Distribution of SRL strategies by game genre

	Fantasy/ fiction	Real-life related storyline (e.g., home, school)	Content area related storyline	Mystery	War	Non-specified
Rehearsal	0	0	1	0	0	2
Seeking information	0	0	0	0	0	1
Elaboration	0	0	0	1	0	0
Organization	0	0	0	0	0	0
Critical Thinking	0	2	0	0	0	1
Planning	0	0	1	0	0	0
Monitoring	1	0	0	0	0	0
Regulating	0	0	0	0	0	1
Time and study environment	0	1	1	0	0	0
Peer learning	0	2	0	0	0	1
Help-Seeking (including online peers, teachers or others)	0	1	0	0	0	0
Self-efficacy	0	4	1	0	0	2
Task value	0	0	0	0	0	0
Goal orientation	0	1	0	0	0	4
Mixed	0	14	5	2	1	4
Total	1	25	9	3	1	16

interactions. These requirements in fact involved students' SRL strategies, such as goal orientation, peer learning, and so on. Tan et al. (2017) developed a serious game for safe administration of blood transfusion which used the flow model including such elements as clear goals, frame story, challenges, focus attention, playability, fun, feedback, and a sense of control. This game integrated such SRL strategies as regulating and goal orientation.

Application domains

As shown in Fig. 7, in higher education, the engineering domain including computer courses had the greatest ratio across all domains (21.82%), followed by social studies including history (18.18%) and language (18.18%), and health, medical, and physical education (14.55%). During the years of 2009 to 2014, game-based learning was sparsely scattered among language, engineering, and non-specified. During the years of 2015 to 2020, domains adopting game-based learning became increasingly diverse, extending to social studies, science, health, medical and physical education, business & management, across disciplines, and mixed domains. In terms of the increased number of game-based learning studies, the greatest growth happened in social studies (10 articles, 18.18%), followed by in health, medical and physical education (8 articles, 14.55%). In terms of percentage of articles in the period of 2015–2020 within 12 years, the distribution was as follows: engineering (8 articles, 14.55%), language (8 articles, 14.55%), science (4 articles, 7.27%), and business or management (4 articles, 7.27%). For example, in medical education, Gauthier et al. (2015) used an online vascular anatomy study aid game to learn vascular structures.

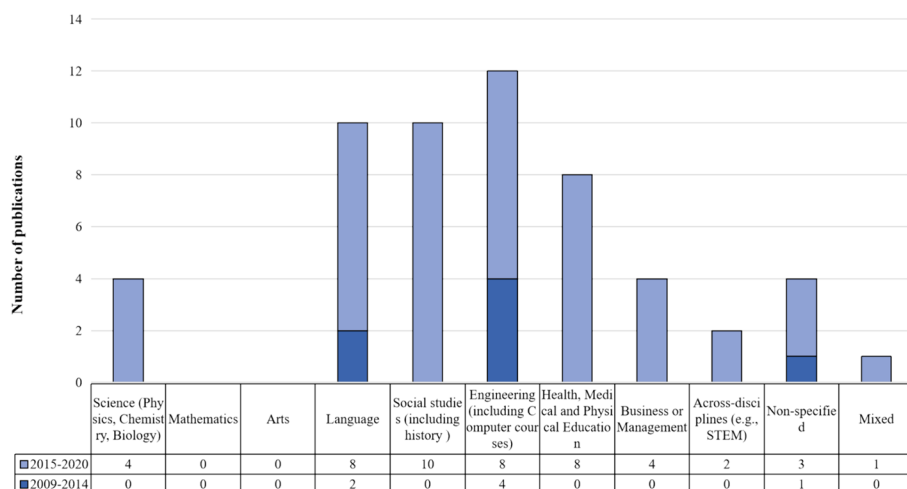


Fig. 7 Distribution of application domains

At the same time, Cozine (2015) adopted a game to enhance learning and facilitate critical thinking within a home-land security curriculum. In the period from 2015 to 2020, playing games was becoming one of the main entertainment approaches for university students with the dramatic development of smartphones (Long et al., 2016). Compared to traditional lecture delivery classrooms in higher education, game-based learning can motivate students to engage in authentic contextual learning (Gómez & Suárez, 2021).

Research methods and duration of these studies

From Table 3, the experimental and mixed research methods make up most research methods, respectively accounting for 43.64% and 36.36% from 2009 to 2020. During the years of 2009 to 2014, only experimental design (3 articles) and mixed research methods (4 articles) were respectively employed in seven studies. Not until the period of 2015 to 2020 did diverse research methods flourish; especially, the experimental design research methods gained the most popular application in 21 studies, followed by mixed methods (16 articles), and surveys (7 articles). For instance, Yeh et al. (2015) designed a true experiment to explore effects of stress on college students' creativity in game-based situations. Proske et al. (2014) also designed a random, controlled true experiment to investigate whether

Table 3 Distribution of research methods

Research methods	2009–2014	2015–2020
Survey	0	7
Experimental design	3	21
Qualitative	0	4
Mixed	4	16

game-based practice outperformed traditional practice in a computer-based writing strategy training program in terms of motivation and achievement.

In terms of research duration in Fig. 8, in the period of 2009 to 2020, studies with a duration of less than 1 day occupied 25.45%, followed by those less than 6 months (20%), those less than 4 weeks (16.36%), those less than 8 weeks (14.55%), those more than 12 months (12.73%), non-specified (9.09%), and those less than 12 months (1.82%). In comparison, research duration during the period of 2009 to 2014 was on average shorter than that during the period of 2015 to 2020. From 2009 to 2014, there were only three published papers with a duration of less than 1 month and only three lasting for more than 2 months. By closely looking at these studies, it was found that studies of less than 1 day focused most on cause-effect analysis (e.g., Biswas et al., 2020; Wu, 2016), followed by learning behavior (e.g., Snow et al., 2015; Tsai et al., 2016). Those studies of less than 4 weeks focused the most on others (e.g., Gordon & Thomas, 2016; Ortiz-Rojas et al., 2019), followed by attitude/motivation (e.g., Chang et al., 2019; Ortiz-Rojas et al., 2019) and learning performance (e.g., Liao & Chen, 2012; Tan et al., 2017), while those studies of more than 4 weeks focused the most on cause & effect analysis (e.g., Buil et al., 2019; Sanchez et al., 2020), and others (e.g., Boldbaatar & Şendurur, 2018; Tao et al., 2015), followed by satisfaction (e.g., Chen et al., 2019a; Venter & Coetzee, 2014), learning performance (Cozine, 2015; Gauthier et al., 2015), and interviews (e.g., Chen et al., 2019a; Ding et al., 2018).

Research issues

As shown in Table 4, research issues related to affect, cognition, skills, learning behavior, correlation or cause-effect analysis, and others were executed to different degrees. In terms of affect dimension, from the years 2009 to 2014, researchers focused most of their attention on satisfaction/interest (57.14%), followed by cognitive load (28.57%) and attitude/efforts (14.29%). Compared to 2009 to 2014, from 2015 to 2020, researchers took great interest in diverse affective aspects involving the technology acceptance model or

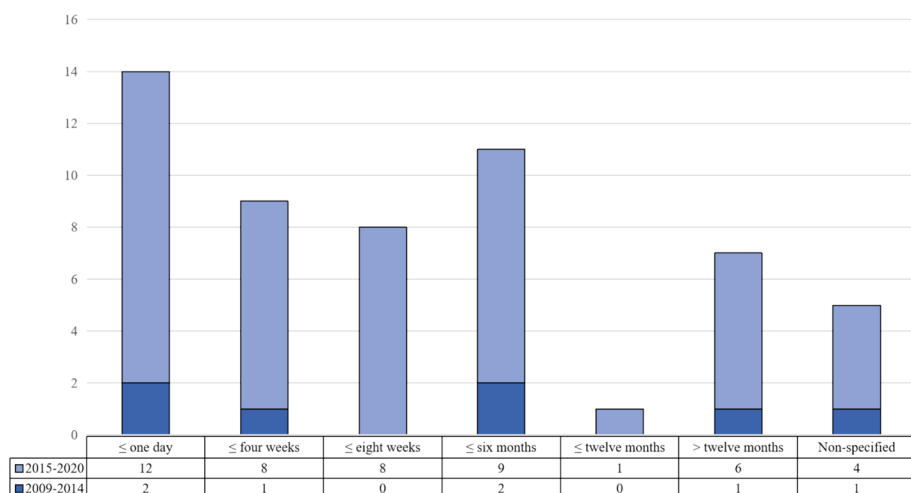


Fig. 8 Distribution of research duration during each period

Table 4 Percentage of research issues in each period

	2009–2014 (<i>n</i> = 7)	2015–2020 (<i>n</i> = 48)	Subtotal (<i>n</i> = 55)
Affect			
Technology acceptance model or intention of use	0%	10.42%	9.09%
Attitudes/efforts	14.29%	29.17%	27.27%
Self-efficacy/confidence	0%	25%	21.82%
Satisfaction/interest	57.14%	27.08%	30.91%
Engagement	0%	12.5%	10.91%
Anxiety	0%	4.17%	3.64%
Cognitive load	28.57%	8.33%	10.91%
Opinions of learners or learning perceptions	0%	25%	21.82%
Cognition			
Learning performance	57.14%	31.25%	34.55%
Higher order thinking skills	14.29%	14.58%	14.55%
Collaboration or communication	0%	2.08%	1.82%
Skills	0%	6.25%	5.45%
Learning behavior	14.29%	25%	23.64%
Correlation or cause-and-effect analysis	14.29%	43.75%	40%
Others	0%	20.83%	18.18%

intention of use, self-efficacy/confidence, engagement, anxiety and opinions of learners or learning perceptions as well as attitudes/efforts, satisfaction/interest and cognitive load. Moreover, studies on opinions of learners/learning perceptions (25%), technology acceptance model or intention of use (10.42%), engagement (12.5%), and anxiety (4.17%) prominently increased. During the years 2015–2020, studies on attitudes/efforts (29.17%), self-efficacy/confidence (25%), satisfaction/interest (27.08%), and opinions of learners/learning perceptions (25%) exceeded 20% as the first rank, the second rank included engagement (12.5%) and the technology acceptance model or intention of use (10.42%) evenly transcending 10%, whereas cognitive load (8.33%) and anxiety (4.17%) were ascribed to the third rank. In terms of cognition, from 2009 to 2014, learning performance (57.14%) was regarded as the main research issue, followed by higher-order thinking skills (14.29%). In contrast, from the years 2015 to 2020, studies on learning performance (31.25%) evidently decreased in comparison with the ratio of those published in the years 2009 to 2014. Contrarily, studies on high-order thinking skills showed a gradual increase in ratio after 2015 (14.58%) compared to before 2015 (14.29%). Moreover, collaboration or communication (2.08%) began to be attended by researchers in the period of 2015–2020. In other words, from 2015 on, researchers started to focus their attention on the social aspects of learning such as collaboration or communication in game-based learning environments. For instance, Chen et al. (2020) explored interdisciplinary experiential learning in an urban (re) development game, which evaluated learners' collaboration and communication. In terms of skills, studies on skills (6.25%) sharply increased during 2015 and 2020 compared to those (0%) in the years of 2009–2014. For example, Topalli and Cagiltay (2018) exploited real-life game development with the Scratch programming environment to improve programming skills during one academic semester period. Besides, in terms of percentage, learning behavior (25%) and correlation or cause-and-effect analysis (43.75%) received

more attention during 2015 and 2020 than in the years 2009 to 2014, which were evenly 14.29%. However, studies attached more importance to correlation or cause-and-effect analysis (43.75%) than learning behavior (25%) during 2015 and 2020. For instance, Stolaki and Economides (2018) investigated the effects of a challenging game within a Facebook group on creativity in an Information Systems undergraduate course using correlation analysis. More studies on other issues (20.83%) were also conducted during 2015 and 2020 compared to the years 2009 to 2014. The holistic data showed that up to 2020, there were still fewer research issues implemented in game-based learning on the technology acceptance model/intention of use, anxiety, collaboration or communication, and skills. Therefore, in the future, such research issues have the potential to be further investigated.

Discussion

This study intended to (1) map trends in research on SRL interwoven with game-based learning environments for higher education, including primary research purpose, SRL strategies, game type and game genre, application domains, research methods and duration of these studies, research issues; (2) identify SRL strategies situated in game-based learning environments as a reference for game developers, teachers and researchers. Self-regulated learning is highly stressed in the sophisticated digital technology era. Game-based learning environments as a context to induce learners' self-regulation have been widely studied over the years. Although many studies have respectively reviewed game-based learning and self-regulated learning, few have been conducted on SRL strategies interwoven with game-based learning. Hence, this research conducted a systematic review of empirical studies that examined SRL situated in game-based learning from the WOS database during 2009 and 2020. In this section, the main findings, limitations and suggestions for future research are discussed as follows.

In general, from 2009 to 2020, the total number of published articles on acquiring skills in game-based learning interwoven with SRL strategies was 2.8 times that of knowledge acquisition and 4 times that of behavior change. From 2009 to 2014, few articles were published on knowledge acquisition, skills acquisition, behavior change, mixed, and others. However, from 2015 to 2020, the total number of published articles on such goals in game-based learning was close to seven times that of from 2009 to 2014. It was also found that SRL strategies embedded in game-based learning exerted a positive effect on learning performance, learning behavior, learning attitudes/effort, self-efficacy/confidence, and satisfaction/interest. The main findings are summarized below.

Primary research purposes

The earliest research purpose of SRL strategies integrated into game-based learning is to gain skills (Sancho et al., 2009), with an increasing number of studies on interweaving SRL strategies with game-based learning for knowledge acquisition (Liu & Chen, 2012; Rodríguez Corral et al., 2014), skills acquisition (Chen & Hsu, 2020; Lee, 2019; Stolaki & Economides, 2018), and behavior change (Hardy & Totman, 2017; Hsu et al., 2018; Tsai et al., 2016). These results are consistent with the argument contended by Zhonggen (2019) that players cultivate their knowledge and practice their skills through overcoming numerous hindrances during gaming. In addition, the research purposes reviewed by this study are akin to the conclusions drawn by Clark et al. (2016) in their meta-analysis of

game-based learning which found that digital games are helpful for conceptual and epistemological understanding (i.e., knowledge acquisition), process skills and practices (i.e., skills acquisition), and player's engagement (i.e., behavior change).

SRL strategies

Most studies focused their attention on mixed strategies with more than two tactics (Chang et al., 2020; Fu et al., 2019), where the greatest number of studies involved goal orientation (Tan et al., 2017; Yeh et al., 2015). Especially, goal setting or goal orientation results in successful goal attainment, which is congruent with the finding concluded by Kizilcec et al. (2017). Such SRL strategies as peer learning (Gonzalez et al., 2014), regulating (Chen et al., 2019b; Ding et al., 2018), planning (Khatri et al., 2019; Topalli & Cagiltay, 2018), seeking information (Biswas et al., 2020; Mavridis & Tsiatsos, 2017), critical thinking (Bawa et al., 2018; Chang et al., 2020), self-efficacy (Buil et al., 2019), and monitoring (Hsu et al., 2018) were moderately attended to by researchers. In some sense, this finding coincides with the argument that metacognition, effort regulation and critical thinking were positively correlated with academic outcomes, and peer learning had a moderate effect on academic outcomes in online higher education learning environments (Broadbent & Poon, 2015). In contrast, task value, elaboration, organization, time and study environment, rehearsal, and help-seeking were less studied by scholars. In other words, cognitive strategies (i.e., rehearsal, organization and elaboration), resource management (i.e., time and study environment, and help seeking) and task value as one of the motivational beliefs were scarcely studied in game-based learning environments for higher education.

Game types and game genres

In terms of SRL strategies interwoven with game-based learning, problem-solving games were the most popular game type. The second most popular game type was simulation, which aligns with the finding of Noroozi et al. (2020). These choices might have been affected by active learning and experiential learning theories as well as learners' preference for games (Acquah & Katz, 2020) since these types of games provide gratification for learners (Sjöblom et al., 2017). Moreover, the majority of problem-solving games and simulation games were designed based on real-life related storylines, followed by not-specified game genres. In such game genres, mixed SRL strategies were most adopted, which showed that learning is a dynamic and cyclic system involving motivational, cognitive or metacognitive, and behavioral processes.

Application domains

Engineering is regarded as the first application domain of integrating SRL strategies into game-based learning, followed by language during the years of 2009 to 2014. Up to 2015, SRL strategies interwoven with game-based learning were firstly exploited in health, medical and physical education domains, business & management, and social studies. In 2016, SRL strategies were adopted in the science domain for game-based learning. In 2018, SRL strategies were firstly adopted in the STEM learning domain for game-based learning. Such an evolution of application domains of SRL in game-based learning aligns with the arguments that game-based learning mainly acquires complex skills and problem-solving

competence in STEM, and promotes transferring skills to authentic learning tasks (Plass et al., 2020).

Research methods and duration

The first greatest number of adopted research methods is experimental design, which is consistent with the findings of Gul and Sozibilir's (2016) research. They mainly examined learning performance, including evaluation of artifacts and test/examination. The second greatest number was mixed methods involving more than two methods, in which qualitative methods, surveys, learning analytic technology, sequential analysis, system development, and experimental design were used. Most studies lasted less than 8 weeks. This finding is congruent with Xia and Zhong (2018). Considering the internal validity of experimental design, in order to control the growth effect, researchers generally prefer to conduct an experiment during a short-term period for observing behavior change, motivational outcomes or effects of treatments on cognition or metacognition. Only one third of the studies published during 2009 to 2020 lasted for over 2 months.

Research issues

The available previous studies explored affect, cognition, and skills. Such a classification is congruent with that in these studies (Tang et al., 2022; Wang et al., 2018). However, more attention was paid to attitudes/efforts, self-efficacy/confidence, and satisfaction/interest in terms of affective aspects, which coincides with other systematic reviews where affective aspects were also focused on by the majority of studies (Tang et al., 2022; Wang et al., 2018), because new media or methods could change affective aspects more easily than cognitive aspects in the short term. In terms of cognitive aspects, learning performance was the first attended focus. As for skills, learning behavior received increasing attention from researchers. However, fewer studies explored anxiety, collaboration or communication, and skills. Since game-based learning aims to help learners to acquire skills and collaboration & communication competences, more studies need to be investigated on effects of GBL on skill acquisition and collaboration & communication.

Despite its useful findings, there are some limitations in this study. Firstly, this review focused primarily on the published articles pertinent to SRL strategies interwoven with game-based learning in higher education, and conducted analyses based on the pre-established coding scheme. Hence, some results might not be highlighted in this analysis. Secondly, since this study used the categorization of SRL strategies given by Anthonysamy et al. (2020) based on Efklides' (2011) and Pintrich's (2000) models rather than on that of Hadwin et al. (2011), socially shared regulation in SRL, which is important for complex GBL, has not been taken into account; thus, the results might not cover all the strategies in complex GBL. Thirdly, since game types and game genres are evolving along with the development of the game industry, this study could not specify current new game types and game genres. Fourthly, considering the authority and reliability of the WOS database, this systematic review only used one database rather than all databases, which might have affected the distribution of papers by year.

According to the findings, this study offers some suggestions for future studies when integrating SRL strategies into game-based learning in higher education.

- (1). There are large openings for future studies to conduct behavior change verification through game-based learning, along with enhancing studies on effects of SRL strategies on learning behavior.
- (2). To date, there have been no related studies covering all strategies of SRL. It is therefore essential to explore effects of all SRL strategies together in a study on research issues such as learning performance, learning behavior, self-efficacy/confidence, satisfaction/interest, attitudes/effort, and so on. In addition, this study has not found that such SRL strategies as organization and time & study environment were applied to game-based learning in higher education. Therefore, in the future it is inevitable to integrate organization or time and study environments into game-based learning in higher education.
- (3). Since the action game type has not been found to be used in game-based learning for higher education, it is suggested that in the future, how to design action games integrating SRL strategies for higher education would be worth exploring. Besides, competitive and strategy games are developed to engage students in SRL. Meanwhile, games developed on the basis of space exploration, civilization/history, and fantasy/fiction are expected to improve students' participation in SRL in higher education.
- (4). With respect to application domains of higher education, it is suggested that in the future, researchers working in the areas of science, mathematics, arts, and across disciplines (including STEM) should attempt to embed SRL strategies into game-based learning. In other words, these domains have many gaps to be filled by applying SRL strategies to game-based learning.
- (5). According to the existing studies, experimental design was the main research method for most studies, with few qualitative studies including interviews being used for deep analysis of learning processes. Hence, it is suggested that future studies should expand research methods to qualitative methods to deeply understand the fine-grained learning process. Besides, since most studies were conducted for less than 6 months, it is suggested that future studies should prolong the experimental duration to two semesters, that is, less than 12 months, in order to carefully investigate the learning growth path.
- (6). With regard to research issues, studies should be conducted on cognitive load, anxiety, the technology acceptance model, higher-order thinking skills, and collaboration & communication affected by SRL strategies in game-based learning in higher education.

Conclusion

This study aimed to propose potential directions for future research by reviewing the articles on SRL strategies interwoven with game-based learning in higher education published from 2009 to 2020. Although previous studies have discussed the benefits and research trends of game-based learning, this study advances the research by

offering SRL strategies integrated into game-based learning for future research. This review found that researchers focus more on effects of different SRL strategies on game-based learning outcomes with the intention to find more effective game-based learning approaches. It was also found that some seldom-discussed issues for game-based learning might deserve a fresh look. According to the previous studies, mixed strategies involving more than two tactics were mostly adopted in game-based learning, in which goal orientation in the mixed strategies was widely involved, whereas task value, elaboration, organization, time and study environment, rehearsal, and help-seeking were less attended by researchers. In addition, this study found that mixed SRL strategies were widely interwoven with problem-solving and simulation games rather than action games or competitive and strategic games. Meanwhile, the experimental research methods were mostly utilized in the extant studies. The number of studies has also increased on SRL strategies interwoven with game-based learning for higher education.

According to the results, some suggestions are proposed for novice researchers or for those researchers who intend to improve their game-based learning studies. The research of SRL strategies interwoven with game-based learning has shown rapid development, indicating that future studies need to focus more attention on how to effectively use SRL strategies in game-based learning in higher education. Studies comparing game-based learning with other conventional learning approaches are not encouraged. On the other hand, some rarely discussed disciplines using SRL integrated with game-based learning (e.g., science, mathematics, arts, STEM), topical educational issues (e.g., Higher-order thinking, collaboration & communication) or qualitative research methods should be encouraged. In addition, comprehensive analysis methods can be used to analyze learners' performance from different perspectives (e.g., skill performance, self-regulated learning process patterns, affect domains, etc.).

Appendix

See Tables 5, 6, 7 and 8.

Table 5 The description for primary research purposes in game-based learning (Abdul Jabbar & Felicia, 2015)

GBL purposes	Description
Acquisition of knowledge	Retention and appreciation of concepts, principles, theorems, laws, etc. by means of searching, browsing, scanning and skimming, observing visuals or intensive reading in GBL environments
Acquisition of skills	Learning skills (analyzing, synthesizing, summarizing, and inferring), subject area skills and problem-solving skills acquired from GBL environments
Behavior change	Learner's regulation of learning through reflection on what he or she has learned
Mixed	More than two purposes as above
other	Motivational outcomes except the aforementioned purposes, for example, learning perception, interest, curiosity, readiness, etc

Table 6 The description for SRL strategies (Anthonyamy et al., 2020; Garcia et al., 2018)

No	SRL strategies	Description	Examples
1	Rehearsal	Activation of information in working memory rather than acquiring new information in long term memory (i.e., memorizing material by overt or covert practice)	(van Ginkel et al., 2019) Fostering oral presentation competence through a virtual reality-based task for delivering feedback
2	Seeking information	The ability to secure further task information from nonsocial sources when undertaking an assignment	(Mavridis & Tsiatsos, 2017) Game-based assessment: investigating the impact on test anxiety and exam performance
3	Elaboration	The ability to connect prior knowledge with new information with the objective of remembering the new material	(Lee, 2019) Her Story or their own stories? Digital game-based learning, student creativity, and creative writing
4	Organization	The ability of a learner to select the appropriate information and organize their thoughts during a learning process	(Yeh et al., 2015) How stress influences creativity in game-based situations: Analysis of stress hormones, negative emotions, and working memory
5	Critical Thinking	The ability of synthesizing and evaluating online material to make them more meaningful and memorable	(Chang et al., 2019)
6	Planning	Activities include skimming material in GBL before playing, doing a task analysis of the problem, planning the sequence, timing, and completions of activities directed at learning goals	(Khatri et al., 2019) Towards mapping competencies through learning analytics: real-time competency assessment for career direction through interactive simulation
7	Monitoring	Recording events, activities or the processes during game-based learning in mind or in e-tools, evaluating and judging whether activities are directed towards the planned goals	(Hsu et al., 2018) Visual behavior and self-efficacy of game playing: an eye movement analysis
8	Regulating	According to monitoring activities, regulating one's own strategies, orientation, behavior, emotion, efforts etc	(Petrovic-Dziedz, 2019) Gamifying Online Tests to Promote Retrieval-Based Learning
9	Time and Study Environment	The capability to manage one's own study time and tasks. Time management skills include time planning, scheduling, and managing one's time	(Tsai et al., 2016) Visual behavior, flow and achievement in game-based learning
10	Peer learning	The collaboration with other students or peers in order to aid in the learning process	(Sancho et al., 2009) Adaptive Role Playing Games: An Immersive Approach for Problem Based Learning
11	Help seeking	Asking other people for help, such as the instructor or one's peers, or consulting external help and resources	(Lin et al., 2020)
12	Self-efficacy	The capability of a student to confidently navigate through game-based learning environments and produce a positive outcome	(Wu, 2016) Gamification design: A comparison of four m-learning courses

Table 6 (continued)

No	SRL strategies	Description	Examples
13	Task value	An individual's beliefs about the value of doing a task, comprising the sum of the components of attainment value, utility value, and intrinsic value, minus the cost value component	(Cho & Castañeda, 2019) Motivational and affective engagement in learning Spanish with a mobile application
14	Goal orientation	General goals formulated by a learner to a course as a whole, including three general orientations which are mastery, extrinsic, and relative ability	(Sanchez et al., 2020) Gamification in the classroom: Examining the impact of gamified quizzes on student learning
15	Mixed	More than two strategies as above	(Biswas et al., 2020) Multilevel Learner Modeling in Training Environments for Complex Decision Making

Table 7 The description or examples for game types and game genres (Abdul Jabbar & Felicia, 2015; Grace, 2005)

Game Types	Description
Role playing (RPG)	Games that offer the player an opportunity to immerse themselves in the player character's situation
Puzzle-based	Games that offer puzzles as the primary attraction to games. These games are most commonly released on low budgets via the web
Action	Games that offer intensity of action as the primary attraction
Competitive	Games that offer a competitive mechanism to motivate players to play motion-based games
Simulation	The primary gameplay element of a simulation is its ability to match real-world situations
Problem solving	Games that offer exploration and puzzle solving as the main attraction
Strategy	Strategy games entertain through reasoning and problem solving
Multitype	More than two types as above
Game genres	Description/Example
Fantasy/fiction	Kingdom hearts, Fable/Doom, Half life
Real-life related storyline (e.g., home, school)	Max payne
Content area related storyline	Dokific
Space exploration	Hello games
Civilization/history	Second Life
Crime	Grand theft auto
Mystery	Indigo prophecy
War	Metal gear solid, Ghost recon
Non-specified	Not ascribed to any genre as above

Table 8 The description or example for research methods (Lai, 2020)

Research methods	Description or Example
Survey	Descriptive or survey studies
Experimental design	Pre-, quasi-or true experimental design research
Qualitative	Interview or content analysis research
Mixed	Including experimental design/survey and qualitative studies

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Data availability The data of this study can be provided upon sending requests to the corresponding author.

Declarations

Conflict of interest The authors would like to declare that there is no conflict of interest in this study.

Consent to participants This study did not need participants.

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